

Suggested Teaching Guidelines for
**Concepts of Operating System and
Software Development Methodologies**
PGCP-AC, August 2026

Total Duration: 60 Hours: 40 Theory (T) + 20 Lab (L)

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment – 20%

I. Concepts of Operating System

Duration: 22 Hours: 16 Theory (T) + 06 Lab (L)

Objective: To provide students with a fundamental understanding of operating system concepts, Linux environment, process and memory management, virtual memory, and synchronization mechanisms.

Prerequisites: Basic knowledge of computer organization and programming fundamentals.

List of Course Material

Text Book:

No specific courseware for the module, faculty may share some course materials.

Reference book:

1. Silberschatz's Operating System Concepts, Global edition, Abraham Silberschatz, Peter B. Galvin, Greg Gagne/ Wiley

Session 1: Introduction to OS (2T)

- Introduction to Operating Systems
- Role and Functions of an Operating System
- Operating System as System Software
- Hardware Dependency of Operating Systems
- Different components of OS
- Functions, Types and Examples of OS
- User and Kernel space and mode
- Interrupts and system calls

Session 2: Introduction to Linux (2T+ 2L)

- Working basics of file system
- Commands associated with files/directories & other basic commands
- Operators like redirection, pipe
- File permissions
- Permissions (chmod, chown, etc); access control list
- System variables like PS1, PS2
- Shell Programming
 - Introduction to Shells and Different types of Shell in Linux

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- Shell variables; Wildcard symbols
- Shell meta characters; Command line arguments; Read, Echo

Lab:

- Working with various OS commands
- Shell programs related to Session 2

Session 3: Shell Programming (2T+ 4L)

- Decision loops (if else, test, nested if else, case controls, while...until, for)
- Regular expressions; Arithmetic expressions
- Examples in Shell Programming

Lab:

- Shell Programs related to Session 3

Sessions 4 & 5: Processes (4T)

- Definition of a Process
- Preemptive and non-preemptive processes
- Difference between a process and thread
- Process Management
- Process Life Cycle
- Schedulers and Types of Schedulers
- Process Scheduling Algorithms
 - FCFS
 - Shortest Job First
 - Priority
 - Round Robin
 - Queue
- Belady's Anomaly
- Scheduling algorithms to find turnaround time to find the better performing scheduler.
- Process creation using fork; waitpid and exec system calls
- Examples on process creation
- Parent and child processes
- Orphan and zombie processes

Sessions 6 & 7: Memory Management (4T)

- Definition and Types of Memory
- Introduction to Memory Management
- Need of Memory Management
- Continuous and Dynamic allocation
- First Fit, Best Fit, Worst Fit

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- Compaction
- Internal and External Fragmentation
- Segmentation
 - Introduction to Segmentation
 - Hardware requirement for segmentation
 - Segmentation table and its interpretation
- Paging
 - Introduction to Paging
 - Hardware required for paging
 - Paging table
 - Translation look aside buffer
- Concept of Dirty bit
- Shared pages and reentrant code
- Throttling
- IO management

Session 8: Virtual memory and Deadlock (2T)

- Virtual Memory
 - Introduction to Virtual memory
 - Demand paging
 - Page faults
 - Page replacement algorithms
- Deadlock
 - Necessary conditions of deadlock
 - Deadlock prevention and avoidance
 - Semaphore
 - Mutex
 - Producer Consumer problem
 - Dead-lock vs Starvation

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II. Software Development Methodologies

Duration: 38 Hours: 24 Theory (T) + 14 Lab (L)

Objective: To provide practical exposure to Software Development Methodologies, Agile, DevOps and CI/CD practices used in modern application development.

Prerequisites: Basic knowledge of web technologies and web application development.

List of Course Material

Text Book:

No specific courseware for the module, faculty may share some course materials.

Reference book:

1. Software Engineering: A Practitioner's Approach, 9th Edition, Roger S. Pressman, Bruce R. Maxim / McGraw Hill
2. Pro Git, 2nd Edition, Scott Chacon, Ben Straub / O'Reilly
3. Docker Deep Dive, 2020 edition, Nigel Poulton / O'Reilly

Sessions 1 & 2: Git (4T + 4L)

- Introduction to Version Control Systems
- Evolution of Version Control Systems
- Introduction to Git
- Git repository structure: Working directory, Staging area, Local Repository
- Basic Git Operations: git init, git add, git commit, git status, git log
- Branching and Merging
 - Creating, Switching and Merging branches
 - Resolving merge conflicts
- Introduction to GitHub
 - Creating a Repository
 - Cloning a Repository
 - Pushing and Pulling Changes

Lab:

- Create a local Git repository
- Add files and commit changes to the repository
- Modify existing files and track changes
- Use Git commands to:
 - Check repository status
 - View commit history

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- Compare file versions
- Create and switch branches
- Merge branches
- Resolve merge conflicts
- Create a GitHub repository
- Push a local repository to GitHub
- Clone a repository from GitHub

Sessions 3 & 4: Software Engineering (4T)

- Introduction to Software Engineering
 - Software Process
 - Software Process Model
 - Software Product
- Importance of Software Engineering
- Software Development Life Cycles
- Requirements Engineering
 - Types of Requirements
 - Steps involved in Requirements Engineering
 - Requirement Analysis Modelling
- Design and Architectural Engineering
 - Characteristics of Good Design
 - Function Oriented vs Object Oriented System
 - Modularity, Cohesion, Coupling, Layering
 - Design Models
 - UML
- Coding
 - Programming Principles
 - Coding Conventions
- Object Oriented Analysis and Design

Sessions 5 & 6: Agile Development (4T + 2L)

- Introduction to Agile development model
- Agile development components
- Benefits of Agile
- Tools used for agile web development
- Scrum and Extreme Programming
- Introduction to Atlassian Jira
 - Add Project
 - Add Tasks and sub-tasks

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- Create sprints with tasks
- Case study of developing web application using agile methodology

Lab:

- Create a Jira account and log in to Jira Cloud.
- Create Epics, User Stories, Tasks, and Sub-tasks.
- Assign tasks to team members and set priorities.
- Create and configure a Sprint.
- Add backlog items to the Sprint.
- Start and manage the Sprint.
- Update task status using the Scrum/Kanban board.
- Track Sprint progress using Jira reports and dashboards.

Sessions 7 & 8: DevOps (4T + 4L)

- Introduction to DevOps
- DevOps ecosystem
- DevOps phases
- Introduction to containerization
- Docker Overview
- Creating docker images using Docker file
- Container life cycle
- Introduction to YAML
- Introduction to Docker Swarm and Docker Stack

Lab:

- Install and configure docker
- Create docker image using Docker file
- Start docker container
- Connect to docker container
- Copy the website code to the container
- Use docker management commands to
 - List the images
 - List the containers
 - Start and stop container
 - Remove container and image

Session 9: Orchestration (2T)

- Introduction to Kubernetes
- Creating service in Kubernetes
- Configure Kubernetes

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- Configure Kubernetes Dashboard
- Setup a Kubernetes cluster
- Access application using Kubernetes service
- Deploy the website using Dashboard

Session 10: CI/CD Pipeline (2T + 2L)

- Introduction to Delivery Pipeline
- Introduction to Jenkins
- Jenkins management
- Adding slave node to Jenkins
- Building a delivery pipeline

Lab:

- Install and configure Jenkins
- Build a pipeline job using Jenkins

Session 11: Software Testing Overview (2T)

- Introduction to Software Testing
- Verification and validation
- Quality Assurance vs Quality Control vs Testing
- Introduction to STLC and V Model
- Types of testing: Manual and Automation
- Introduction to testing methods: white-box, black-box and grey-box
- Introduction to functional and non-functional testing

Session 12: Selenium (2T + 2L)

- Overview of Automation Testing tools
- Introduction to Selenium (use Eclipse IDE)
- Load web driver
- Create selenese commands: locators: by ID, name, class, tag name, XPath
- Add interactions: text box, radio button selection, check box selection, drop down item selection, keyboard actions, mouse actions, multi select

Lab:

- Create a test plan for project and document the use cases
- Create test case document for different sprints (designed in SE)
- Create a test suite using Selenium
 - Add commands and interactions